# UNIT-2
# EFFECTIVE SPEAKING

## TOPIC-2
## ORGANS OF SPEECH

# ORGANS OF SPEECH

- **CONTENTS:**
- **1.The Respiratory System**
- **2.The Phonatory System**
- **3.The Articulatory System**

# Organs of Speech

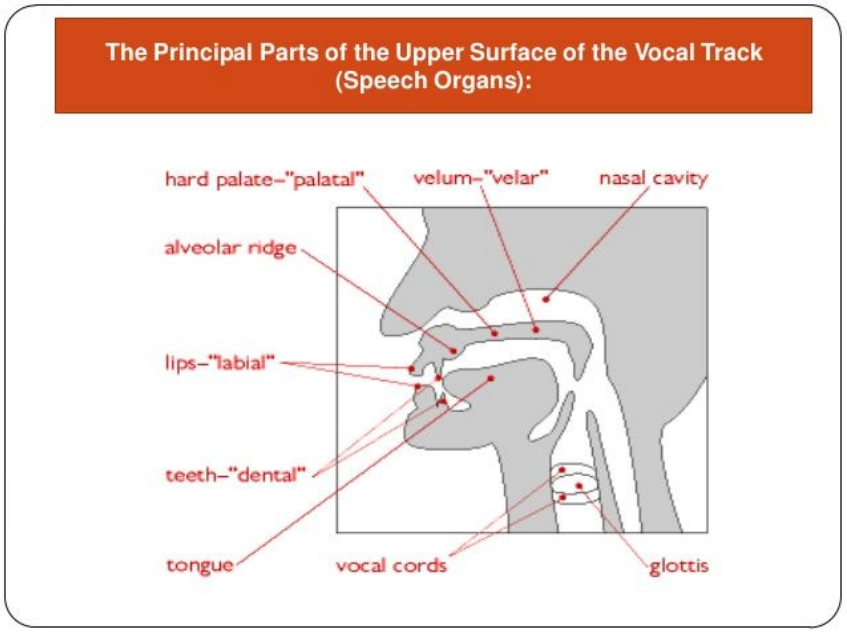
- **1.Respiratory System**: Comprises of lungs, the muscle of chest and the windpipe(trachea)
- **2.The Phonatory System**: Comprises of **Larynx**
- **3.The Articulatory System:** Comprises of nose,the teeth, the tongue, the roof of the mouth and the lips.

# The Respiratory System

- The **respiratory system** consist of the **lungs,the muscles of the chest** and the **wind pipe(** also called as **trachea).**
- The lungs are spongy bodies. They are made up of small sacs called **alveoli.**
- It is in these sacs that the blood is cleaned and is provided with fresh oxygen from the outer-air.
- Air is supplied to Alveoli by small tubes called the bronchioles.
- The bronchioles come together into two large tubes called bronchi,one situated on the right and the other on the left.
- The bronchi join the trachea or the windpipe.
- It is through the trachea that the air that we breathe in passes through the throat into the lungs.
- The act that is commonly known **as respiration involves two process- taking outer air into the lungs( called inspiration**) and **throwing out air from the lungs into the outer atmosphere ( called expiration)**

# 2.The Phonatory System

- The larynx is also called **as 'Adam's Apple' ,** situated at the top of the windpipe.
- The air from the lungs has to come out through the windpipe and the Larynx.
- In the larynx are situated in pair of lip like structure.
- These are called the **Vocal cords** and these are placed horizontally from front to back.
- They are attached in front and can be separated at the back.
- The opening between the cords is called **the glottis.**

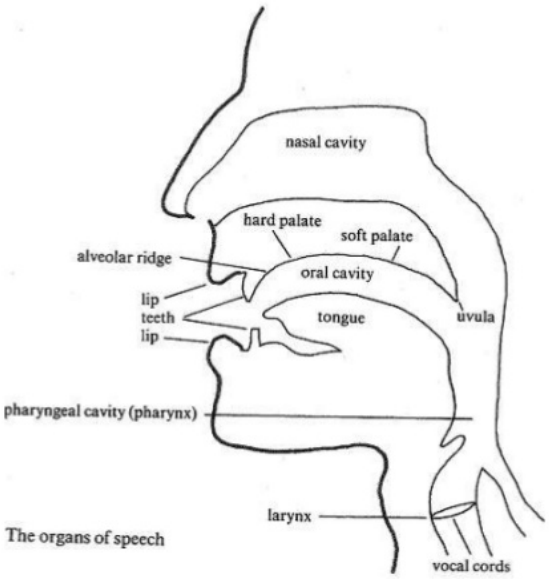


# 2.The Phonatory System

- The vocal cords can be opened and closed and when the two cords come very close and when the two cords very close to each other, the glottis will be shut completely.
- In fact when we swallow food and water, the vocal cords shut the glottis and thus prevent the food or water from entering the windpipe.
- When we breathe in and out, the vocal cords are drawn wide apart and thus the glottis is open
- The air enters the lungs or get out through the wide open glottis.
- When we produce some speech sounds, the **vocal cords are wide apart and the glottis is open**.
- **Such sounds produced with a wide –open glottis and are called as voiceless.** Sounds. The first sounds in **English word peel, ten, keen, chin, fine, seen, shine and hat are voiceless sounds.**

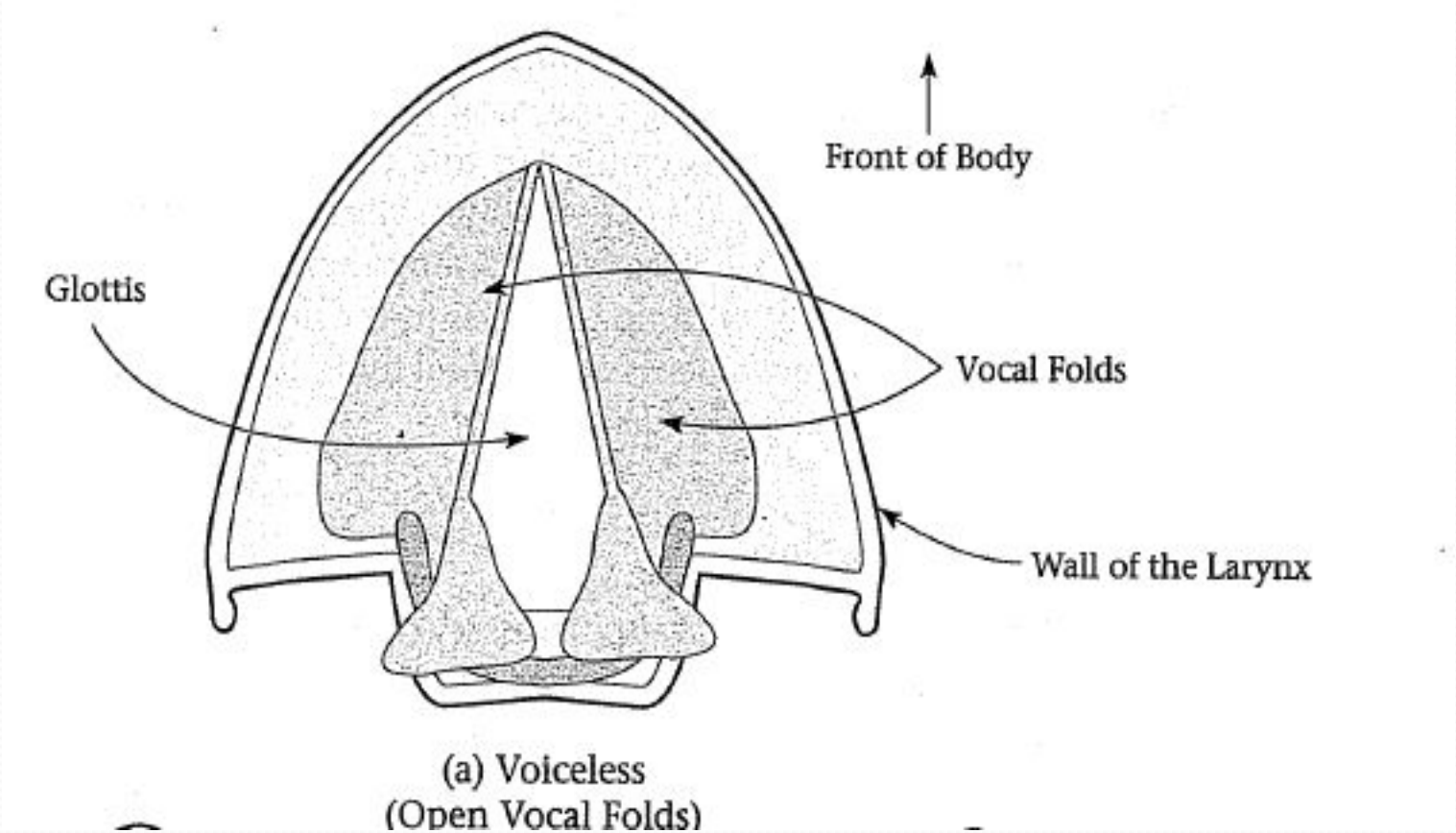
# 2.The Phonatory System(cont…)

- During the production of certain sounds, the vocal cords **<u>are loosely held together</u>** and the pressure of the air from the lungs makes them open and close rapidly.
- This is called the **vibration of the vocal cords and the sounds produced when the vocal cords vibrate are called <u>voiced sounds</u>.**
- **All the sounds in English words bead, deed, girl, judge, vine, measure, need , wing, red, yard and well <u>are voiced sounds.</u>**

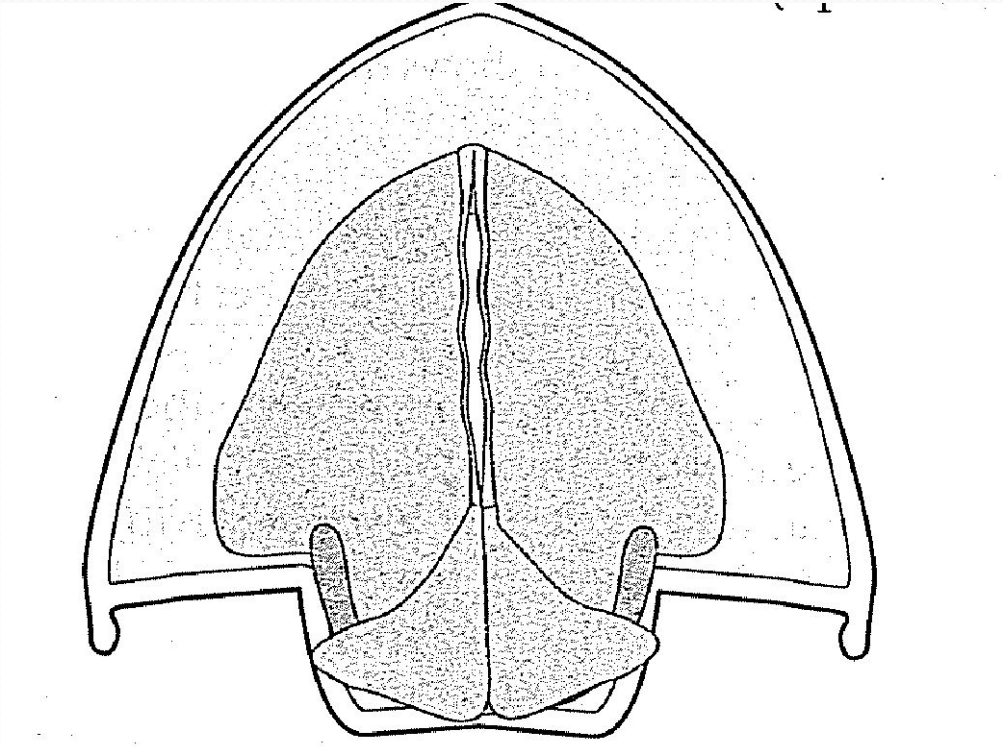


The organs of speech

# Voicing


Front of Body
Glottis
Vocal Folds
Wall of the Larynx
(a) Voiceless
(Open Vocal Folds)

# Voicing



(b) Voiced
(Approximated Vocal Folds)

# III- The Articulatory System

- After Larynx, the airstream passes through other 9 organs of speech that are laying above the larynx.

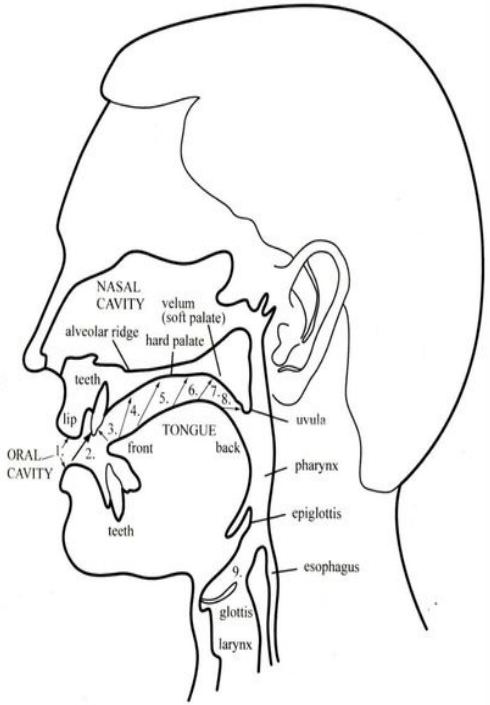
**1 – The Lips** **2- The Teeth**

**3- The Alveolar ridge** **4-The hard palat**

**5- The Soft Palate** **6- The Uvula**

**7- The tongue** **8- The Pharynx**

**9- The glottis**




- These Organs of speech make **the articulatory System** together.

# 1 – The Lips

The upper and lower lips have important role in producing certain speech sounds especially in producing vowels .

Movements:

1- Closing the lips and then releasing the closure abruptly EX: **p**ut , **b**oot

2- closing the lips and making nasal consonant EX: **m**en

3- protrude and rounded EX: b**oo**t

4- spread EX: b**ea**ds

5- neutral – not spread not rounded EX: **a**bout

The T

## 2- The Teeth

Certain consonants are produced with the help of the teeth.

Movements:

-Putting the tongue in between the front teeth which are called **interdental fricatives**. EX: **th**ink, **th**at

- Pressing the lower lip with the upper front teeth Which are called **labiodental fricatives**. EX: **f**an, **v**an

# 3- The Alveolar Ridge

It's located just behind the upper front teeth and consonantal sounds like **t, d, n, l, s, z** are produced here.

Movements:

- Tongue tip or blade touches the alveolar ridge. EX: l, n
- making a complete closure by the tip of the tongue and the alveolar ridge, followed by a sudden release of the closure . EX: t, d
- Narrowing the passage of the air at the alveolar ridge and forcing the air out through this passage
- EX: s, z

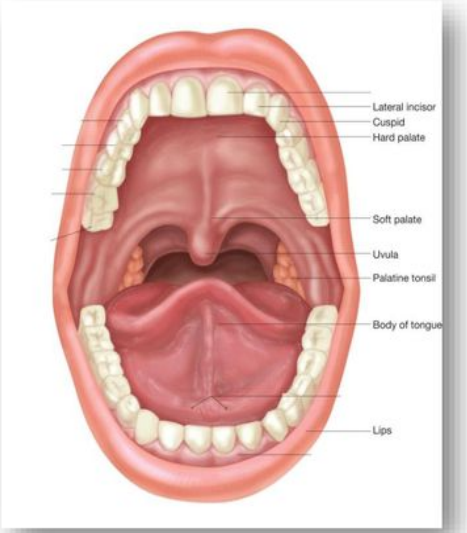
# 4- The Hard Palate

The hard bony surface of the roof of the mouth, immediately after alveolar ridge. It's not moveable.

Movement:

The front part of the tongue moves toward the hard palate.

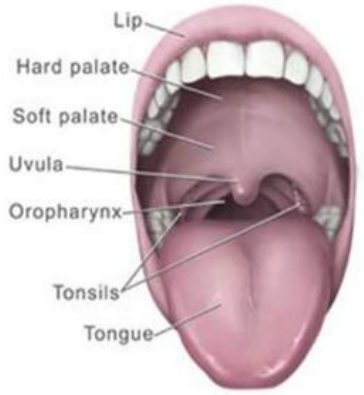
EX: **y**es



Soft P

# 5- The Soft Palate or Velum

The soft and moveable part of the roof of the mouth, immediately after the hard palate.



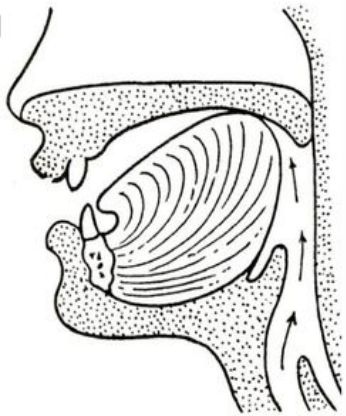

Movements:

- Velum dropped down and the back of the tongue makes a closure with velum which is called **Velar closur**

EX: **k**ing

- Velum raises itself and make a closure with the Back wall of the pharynx which is called **velic closure**.

EX: **g**o



Velar and Velic Closures.

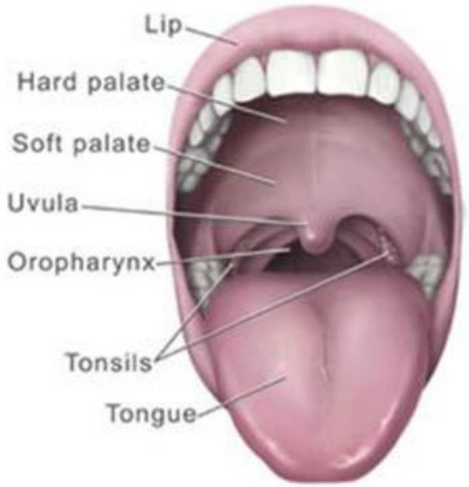
Uvul

## 6- The Uvula

It's a small fleshy pendent structure at the very end of the soft palate.

Movement:

The back of the tongue articulates with uvula and produce **q**-sound.



To

## 7- The Tongue

-It's the most important single organ of speech.

-Even in many languages the word TONGUE means Language too.

-It's flexible and can make a large number of gestures to produce speech sounds.

**- It has different parts:**

1- tip 2- blade 3– front

4– back 5- rims

# Differ

## Different parts of the tongue

1- The Tip: When the tongue is the state of the rest , it's the part that is lying behind the lower front teeth.

2- The Blade: It 's lying the opposite side of the alveolar ridge.

3- the Front: It faces the hard palate.

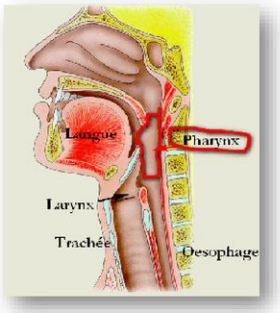
4- The Back: It's on the opposite side of the soft palate.

5- The Rims: They are the edges of the tongue.

# Pharynx

## 8- The Pharynx

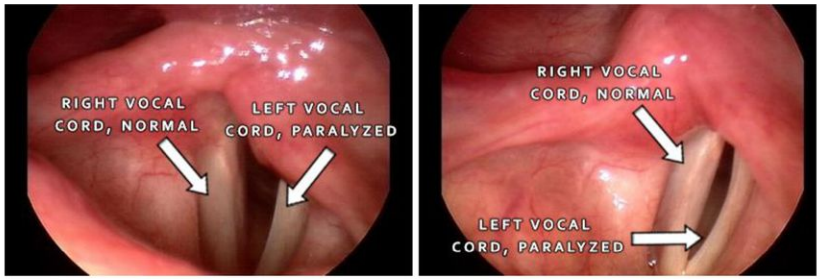
It's like a tube just above the Larynx and before Uvula. It's top is divided in **2** parts: One part goes to oral cavity and another goes to nasal cavity.



# Glottis

## 9- The Glottis

It's The space between the Vocal Cords.




Movement:

With a rapid closing the opening of the glottis, the air stream traps behind it then with a sudden release of the air makes glottal stop sounds.

EX: ma**tt**er, bu**tt**er

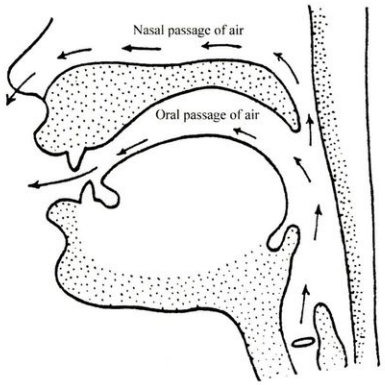
- Oral Cavity: /a/, / e/, /l/, /o/,  /u/, c/,/g,/d/,b
- aim, come, go, see, you

- **Nasal Cavity:**
- /m/, /n/, ing
- Man, moon, no, now, ring, morning

# Oral and Nasal Passage

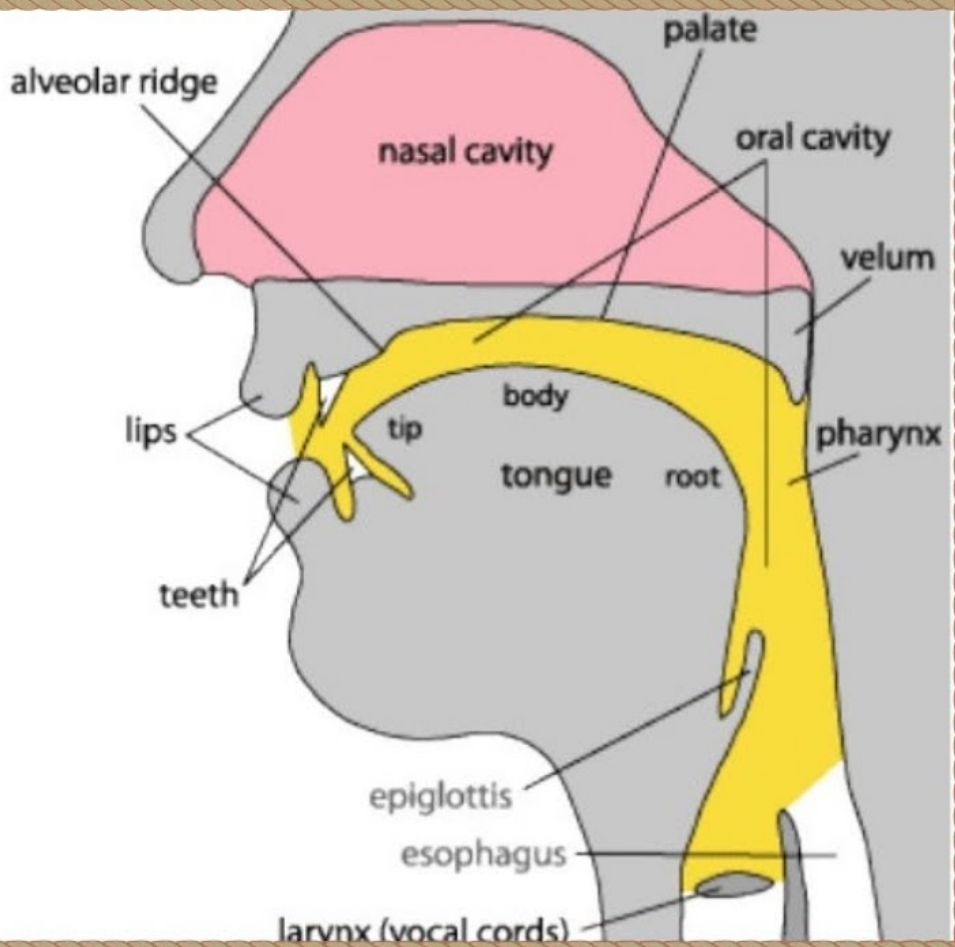
## Oral and Nasal Cavities

The air is pushed by the lungs and travels through the mouth which is called **oral cavity** or through the nose which is called **nasal cavity**.




Nasal Passage Open; Oral Passage Also Open.

# Organ Of Speech & Their Functions



# questions?

- Briefly explain Respiratory system?
- What is Phonatory system ?Explain with the help of diagram.
- Briefly explain Articulatory System with the help of diagram.(Oral Cavity & Nasal Cavity)